World–Action Model | Network Detail

One member · one time step

a Input features

Node input — 45

Position	2
Fourier position features	24
Three displacement vectors	6
History validity flags	3
Shape descriptors	6
Wall gap + direction	3
Image-edge flag	1

Node encoder: 45 → 64 → 64 → 64

Edge input — 5

Relative position	2
Centre distance	1
Estimated surface gap	1
Log-area difference	1

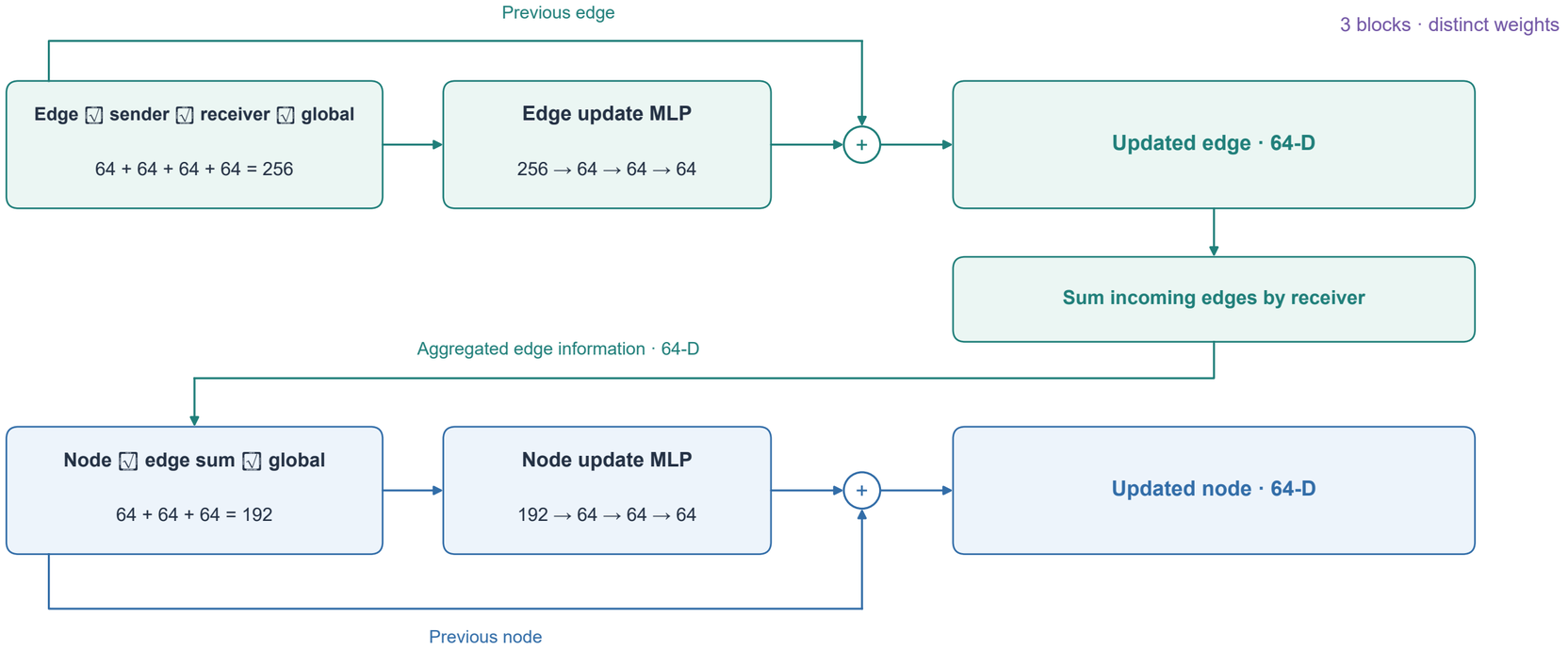
Edge encoder: 5 → 64 → 64 → 64

Global input — 12

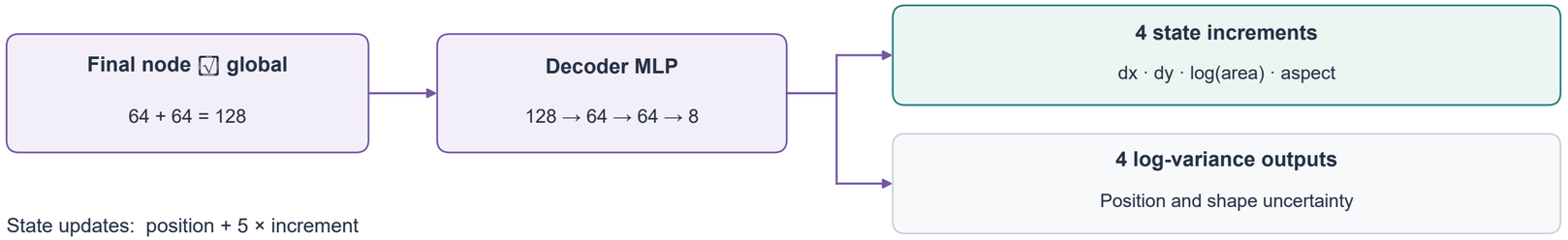
Current + previous PZT amplitudes	8
Time since the last switch	1
Mean dx, dy and displacement length	3

Global encoder: 12 → 64 → 64 → 64

b Residual message-passing block



c Decoder and state update



State updates: position + 5 × increment

log(area), aspect + 0.1 × increment

Every MLP: Linear → SiLU → Linear → SiLU → Linear